

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of B. Pharm. Examination

University Theory Examination OCTOBER 2017

PH117/PH125 HUMAN ANATOMY AND PHYSIOLOGY II

Date: 02.10.2017, MONDAY Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. Which of the following are the parts of neurons?
 - A. brain, spinal cord, and vertebral column
 - B. dendrite, axon, and cell body
 - C. sensory and motor
 - D. cortex, medulla and sheat
 2. A dendrite conducts nerve impulses _____ the cell body.
 - A. away from
 - B. toward
 - C. both toward and away from
 - D. around, bypassing
 3. Gaps in the myelin sheath are called _____.
 - A. nodes of Ranvier
 - B. the synapse
 - C. axonal interstices
 - D. myelinoids
 4. With an action potential, depolarization of the axomembrane is recorded as the gates open, allowing _____ to flow into the axon.
 - A. potassium ions
 - B. H^+ ions
 - C. sodium ions
 - D. all of the above
 5. The _____ nerve carries parasympathetic impulses to visceral muscles.
 - A. vagus
 - B. abducens
 - C. accessory
 - D. trigemina
 6. Which of the following are characteristic of the sympathetic system?
 - A. inhibits the digestive tract
 - B. dilates the bronchi
 - C. accelerates the heartbeat
 - D. all of the above
 7. Where are the posterior pituitary hormones manufactured?
 - A. in neurosecretory cells that originate in the hypothalamus
 - B. in the posterior pituitary
 - C. in the anterior pituitary

- D. in the brain stem
8. Sex hormones are secreted by _____.
 - A. all endocrine tissues of the body
 - B. the inner cortex of the adrenal medulla
 - C. the inner cortex of the adrenal medulla and the gonads
 - D. the gonads
 9. Which of these is not an endocrine gland?
 - A. pancreas
 - B. testes
 - C. salivary gland
 - D. parathyroid
 10. Which hormone has the greatest effect on metabolism?
 - A. GH
 - B. thyroxine
 - C. TSH
 - D. ACTH
 11. Prostaglandins are _____ substances that regulate neighboring cells.
 - A. endocrine
 - B. paracrine
 - C. autocrine
 - D. glycoprotein
 12. Males have a number of internal accessory organs. Which one is responsible for secreting fluid containing fructose and prostaglandins?
 - A. epididymis
 - B. seminal vesicles
 - C. vas deferens
 - D. prostate gland
 13. During ovulation, a(n) _____ is released from the ovary.
 - A. ovum
 - B. secondary oocyte
 - C. primary oocyte
 - D. oogonium
 14. The detection of the hormone _____ is the basis of pregnancy tests.
 - A. human chorionic gonadotropin (hCG)
 - B. follicle-stimulating hormone (FSH)
 - C. luteinizing hormone (LH)
 - D. oxytocin
 15. Which structure is formed following ovulation?
 - A. Graafian follicle
 - B. secondary oocyte
 - C. corpus luteum
 - D. coronal radiata
 16. Which factor favors spermatozoa development?
 - A. temperature of 37°C
 - B. temperature of 34°C
 - C. acidity
 - D. estrogens
 17. Ovulation appears to be triggered by a midcycle surge in _____.
 - A. Estrogen
 - B. progesterone
 - C. follicle-stimulating hormone (FSH)
 - D. luteinizing hormone (LH)

18. The type of joint that is distinguished by having a fluid-filled joint cavity is a _____ joint.
- fibrous
 - cartilaginous
 - synovial
 - suture
19. The shoulder and hip joints are of this type.
- ball-and-socket
 - pivot
 - saddle
 - gliding
20. How many phalanges are present in each hand?
- 28
 - 14
 - 15
 - 30

SECTION – II

- Q 2** Attempt any **TWO** of the following:
- Write a note on synthesis, storage and release on thyroid hormones. **05**
 - Discuss the physiological role of histamine. **05**
 - Classify joints. Discuss any one of them in detail. **05**
- Q 3** Attempt any **TWO** of the following:
- Write a note on adrenal gland. **05**
 - Enumerate the hormones of pancreas and discuss any one of them in detail. **05**
 - Differentiate between sympathetic and parasympathetics nervous systems. **05**

SECTION – III

- Q 4** Attempt any **FOUR** of the following:
- Write note on reflex arc mechanism. **05**
 - Describe the Five main functions of the skeletal system. **05**
 - Explain process of oogenesis. **05**
 - Describe the functions of endocrine system and enlist different endocrine glands and their hormones. **05**
 - Explain the neurohormonal changes in female menstrual cycle. **05**
 - Draw the neat and labeled diagram of brain. **05**
- Q 5** Attempt any **FOUR** of the following:
- Write a note on medulla oblongata. **05**
 - Write note on cranial nerves. **05**
 - Draw the neat and labeled diagram of male reproductive system. **05**
 - Define neurotransmitters. Enumerate neurotransmitters. Write note on any one neurotransmitter of nervous system. **05**
 - Write note on spermatogenesis. **05**
 - Explain the compositions, formation, release and functions of CSF. **05**